

Midpoint Circle Drawing Algorithm

1. Input the center coordinates (x_c, y_c) and radius r
2. Set $x = 0, y = r$
3. Calculate the initial decision parameter $p = 1 - r$
4. Repeat the following steps while $x \leq y$
 - a. Plot the points using 8-way symmetry:
 $(x_c + x, y_c + y), (x_c - x, y_c + y), (x_c + x, y_c - y), (x_c - x, y_c - y),$
 $(x_c + y, y_c + x), (x_c - y, y_c + x), (x_c + y, y_c - x), (x_c - y, y_c - x)$
 - b. Increment x by 1
 - c. If $p < 0$ then
 $p = p + 2x + 1$
else
 $y = y - 1$
 $p = p + 2(x - y) + 1$
5. Repeat until $x > y$
6. Stop